

Drawing 2D Shapes in HTML5 Canvas – BCA Notes

Introduction

HTML5 introduced the **Canvas** element, which allows developers to draw graphics directly on a web page using JavaScript.

Canvas is used for:

- Drawing shapes
- Creating charts and graphs
- Developing games
- Image manipulation
- Animations
- Interactive applications

Syntax

```
<canvas id="myCanvas" width="500" height="300">  
Your browser does not support canvas.  
</canvas>
```

Attributes

Attribute	Description
id	Unique identifier for canvas
width	Width of canvas area
height	Height of canvas area

What is Canvas?

The `<canvas>` element provides a rectangular area where graphics can be drawn using JavaScript.

Example

```
<canvas id="myCanvas" width="500" height="300"></canvas>
```

To draw on the canvas:

```
var canvas = document.getElementById("myCanvas");  
var ctx = canvas.getContext("2d");
```

Explanation

- `getElementById()` selects the canvas.
 - `getContext("2d")` creates a 2D drawing environment.
 - `ctx` is the drawing object.
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Coordinate System

Canvas uses a coordinate system.

(0,0)
+-----> X
|
|
|
|
V
Y

- Top-left corner = (0,0)
- X increases toward right.
- Y increases downward.

Example:

```
ctx.fillRect(50,50,100,80);
```

- `x = 50`
 - `y = 50`
 - `width = 100`
 - `height = 80`
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Drawing a Line

Lines are created using:

- `beginPath()`
- `moveTo()`
- `lineTo()`
- `stroke()`

Syntax

```
ctx.beginPath();  
ctx.moveTo(50,50);  
ctx.lineTo(200,50);  
ctx.stroke();
```

Example

```
<canvas id="c1" width="300" height="200"></canvas>

<script>
var canvas = document.getElementById("c1");
var ctx = canvas.getContext("2d");

ctx.beginPath();
ctx.moveTo(50,50);
ctx.lineTo(250,50);
ctx.stroke();
</script>
```

Output

A straight horizontal line.

Drawing Rectangles

HTML5 provides two methods:

1. Filled Rectangle

```
ctx.fillRect(x,y,width,height);
```

Example:

```
ctx.fillStyle="blue";
ctx.fillRect(50,50,150,100);
```

Output

Blue filled rectangle.

2. Rectangle Border

```
ctx.strokeRect(x,y,width,height);
```

Example

```
ctx.strokeRect(50,50,150,100);
```

Output

Rectangle outline.

3. Clear Rectangle

Removes content.

```
ctx.clearRect(50, 50, 100, 100);
```

Drawing Circles

Circles are drawn using the `arc()` method.

Syntax

```
ctx.arc(x, y, radius, startAngle, endAngle);
```

Example

```
ctx.beginPath();  
ctx.arc(150, 150, 50, 0, 2*Math.PI);  
ctx.stroke();
```

Explanation

Parameter	Meaning
x	Center X
y	Center Y
radius	Radius
0	Starting angle
2π	Complete circle

Drawing Filled Circle

```
ctx.beginPath();  
ctx.arc(150, 150, 50, 0, 2*Math.PI);  
ctx.fillStyle="red";  
ctx.fill();
```

Drawing Arcs

An arc is a part of a circle.

Example

```
ctx.beginPath();  
ctx.arc(150,150,50,0,Math.PI);  
ctx.stroke();
```

Output

Semi-circle.

Drawing Triangles

Triangle is drawn using three points.

Example

```
ctx.beginPath();  
  
ctx.moveTo(150,50);  
ctx.lineTo(100,150);  
ctx.lineTo(200,150);  
  
ctx.closePath();  
ctx.stroke();
```

Output

Triangle.

Drawing Polygons

Polygon is a shape having multiple sides.

Example (Pentagon)

```
ctx.beginPath();  
  
ctx.moveTo(150,50);  
ctx.lineTo(200,100);  
ctx.lineTo(180,170);  
ctx.lineTo(120,170);  
ctx.lineTo(100,100);  
  
ctx.closePath();  
ctx.stroke();
```

Output

Five-sided polygon.

Drawing Ellipse

Syntax

```
ctx.ellipse(  
x,y,  
radiusX,  
radiusY,  
rotation,  
startAngle,  
endAngle  
);
```

Example

```
ctx.beginPath();  
  
ctx.ellipse(  
200,100,  
80,40,  
0,  
0,  
2*Math.PI  
);  
  
ctx.stroke();
```

Output

Ellipse.

Colors in Canvas

Fill Color

```
ctx.fillStyle="green";  
ctx.fillRect(50,50,100,100);
```

Border Color

```
ctx.strokeStyle="red";  
ctx.strokeRect(50,50,100,100);
```

Line Width

Example

```
ctx.lineWidth=5;

ctx.beginPath();
ctx.moveTo(50,50);
ctx.lineTo(200,50);
ctx.stroke();
```

Output

Thick line.

Drawing Text

Canvas can display text.

Filled Text

```
ctx.font="30px Arial";
ctx.fillText("Hello BCA",50,100);
```

Outline Text

```
ctx.font="30px Arial";
ctx.strokeText("Hello BCA",50,100);
```

Drawing Images

Example

```
var img = new Image();

img.onload = function()
{
    ctx.drawImage(img,50,50);
};

img.src="image.jpg";
```

Common Canvas Methods

Method	Purpose
beginPath()	Start drawing

Method	Purpose
closePath()	Close shape
moveTo()	Move cursor
lineTo()	Draw line
stroke()	Draw border
fill()	Fill color
fillRect()	Filled rectangle
strokeRect()	Rectangle outline
clearRect()	Clear area
arc()	Circle/arc
ellipse()	Ellipse
fillText()	Draw text
strokeText()	Outline text
drawImage()	Display image

Advantages of HTML5 Canvas

1. Easy to create graphics.
 2. Supports animations.
 3. No external plugins required.
 4. Fast rendering.
 5. Used in games and graphical applications.
 6. Works in all modern browsers.
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Limitations of Canvas

1. Graphics are pixel-based.
 2. Individual shapes cannot be directly manipulated after drawing.
 3. Requires JavaScript knowledge.
 4. Large complex drawings may affect performance.
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Applications of Canvas

- Online games
- Drawing applications
- Photo editors
- Data visualization
- Animation systems
- Interactive learning tools

- Graphs and charts
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Frequently Asked BCA Exam Questions

Short Questions

1. What is HTML5 Canvas?
2. What is the purpose of `getContext("2d")`?
3. Explain the coordinate system of Canvas.
4. Differentiate between `fillRect()` and `strokeRect()`.
5. What is the use of `beginPath()`?
6. Explain the `arc()` method.
7. What is the purpose of `fillStyle`?
8. Write the syntax of `drawImage()`.

Long Questions

1. Explain HTML5 Canvas with suitable examples.
 2. Describe how to draw lines, rectangles, and circles using Canvas.
 3. Explain the coordinate system used in Canvas.
 4. Discuss various Canvas methods used for drawing shapes.
 5. Explain the advantages and applications of HTML5 Canvas.
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Conclusion

The HTML5 Canvas element is a powerful graphics tool that enables developers to create **2D drawings, shapes, text, images, and animations** directly inside web pages using JavaScript. It is widely used in modern web development for creating interactive and visually appealing applications. This is an important topic for BCA students because it combines **HTML5, JavaScript, graphics programming, and web application development**.